

SPSS 6 ANOVA Analysis of Variance

01/05/2010

In this course

1. What is ANOVA?
2. Descriptive statistics before ANOVA
3. ANOVA Test
 - Is there any significance difference among a group of values?
4. Where exactly is the significance difference?
 - Tukey's HSD
5. Requirement of Using F
 - Can we use ANOVA?
6. Two-way ANOVA
 - What's the effect of 2 factors?

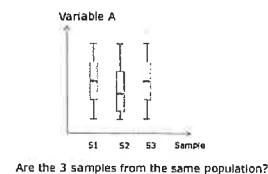
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Scenarios for Statistical Test on Groups

- Scenario 1:
 - You want to find out what is the smallest dosage of a drug that is effective for treating fever
 - Randomly give 5 different dosages to 300 patients with high fever
 - Then record the time that each patient takes to recover
 - Are the 5 drug dosages equally effective?
- Scenario 2:
 - You have 4 possible ways to integrate the components of a product
 - Randomly assign 50 workers to each integration method
 - Then record the defects of each product
 - Are the 4 integration methods equally effective?

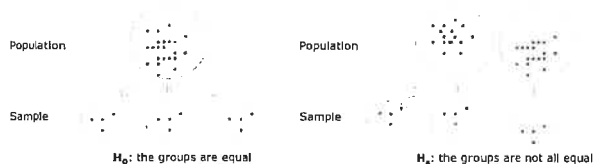
Analysis of Variance (ANOVA)

- ANOVA and F value
 - Measure the differences of multiple independent groups of cases, represented by the F value
 - Each group is a sample from the population
 - The groups are different if they come from different populations
 - Large F value suggests that the groups are different, and come from different populations



Analysis of Variance (ANOVA)

- ANOVA tests the hypothesis of group differences
 1. Null hypothesis H_0 : The groups' means are all equal
 2. Alternate hypothesis H_a : The groups' means are not all equal



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Sources of Variances

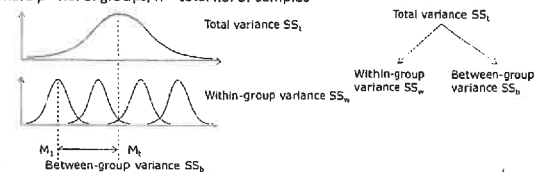
- Sources of variances:
 1. Total variance (SS_t) (SS: Sum of Square)
 2. Variance within groups (SS_w)
 3. Variance between groups (SS_b)

- Relationship of the 3 variances and F test:

$$1. SS_t = SS_b + SS_w$$

$$2. F = \frac{\text{Mean Square}_b}{\text{Mean Square}_w} = \frac{SS_b / (p-1)}{SS_w / (n-p)}$$

where p = no. of groups, n = total no. of samples



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ANOVA Test

- To run ANOVA test:
 1. Use **Analyze > Compare Means > One-way ANOVA >**
 2. Make settings (see red color text below)

Descriptive (Options > Statistic Group > Descriptive)

Variable A (Dependent List)		N	Mean	SD	SE	95% CI for Mean	
Variable B (Factor Variable)						Lower Bound	Upper Bound
81		..	..	..	..	..	..
82		..	..	..	..	..	..
83		..	..	..	..	..	..
Total		..	..	..	..	..	..

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Task 2: ANOVA Test

- Generate One-way ANOVA for the effect of treatment on the change in weight

post-pre weight

3 groups (3 levels) of treatment

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
behavioral treatment	28	3.0099	7.30850	1.35716	Lower Bound	Upper Bound	-8.10	20.90
control	26	-4.5000	7.98870	1.56571	-3.8767	2.7767	-12.20	15.90
family therapy	17	7.2647	7.15742	1.73593	3.5947	10.9447	-5.30	21.50
Total	72	2.7639	7.89360	.94088	8979	4.6399	-12.20	21.50

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Task 2 answer: ANOVA Test

- Steps:
 1. **Analyze > Compare Means > One-way ANOVA**
 2. set **Dependent List: difwt**, **Factor Variable: group**,
 3. **> Options > Statistic Group > Descriptive > Continue>ok**

post-pre weight

3 groups (3 levels) of treatment

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
behavioral treatment	28	3.0099	7.30850	1.35716	Lower Bound	Upper Bound	-8.10	20.90
control	26	-4.5000	7.98870	1.56571	-3.8767	2.7767	-12.20	15.90
family therapy	17	7.2647	7.15742	1.73593	3.5947	10.9447	-5.30	21.50
Total	72	2.7639	7.89360	.94088	8979	4.6399	-12.20	21.50

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ANOVA Test

- Analysis of Variance table
 1. What is the external variance?
 2. What is the internal variance?
 3. What is the conclusion of comparing the 3 sample groups?
 4. How is the confidence on the conclusion?

Small significant level means the chance of getting the F while there is no true difference between group is small, i.e. confident on the F

ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	614.844	2	307.322	5.422	.006
Within Groups	3910.742	69	56.677		
Total	4525.586	71			

Total variance

Large F (> > 1) means there are differences in 3 groups of data.
Recall F = Mean Square_{between} / Mean Square_{within}

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Task 3: Contrasts: testing linear combinations of means

Linear contrast of means: check null hypothesis of $a_1\mu_1 + a_2\mu_2 + \dots + a_n\mu_n = 0$ instead of $\mu_1 - \mu_2 = 0$, where a_i is the contrast coefficient, with $\sum a_i = 0$.

1. **Statistic Tool Bar > Dialog Recall > One-way ANOVA > Post Hoc >**
2. **Contrast > set Coefficients to: 2 > add, -1 > add, -1 > add > Continue > OK** (you may decide the coefficients with reference to the means plot first, or before conducting the test)

Contrast Coefficients

Contrast	behavioral treatment	control	family therapy
1	2	-1	-1

$\sum a_i = 2(\text{Behavioral group mean}) - 1(\text{Control group mean}) - 1(\text{Family group mean}) = -0.8009$

Value of contrast / Std. Error

Contrast	Value of Contrast	Std. Error	t	df	Sig. (2-tailed)
post-pre weight	-8.009	3.65122	-2.19	69	.027
Does not assume equal variances	-8.009	3.59269	-2.24	59.972	.024

>0.05, accept the null hypothesis

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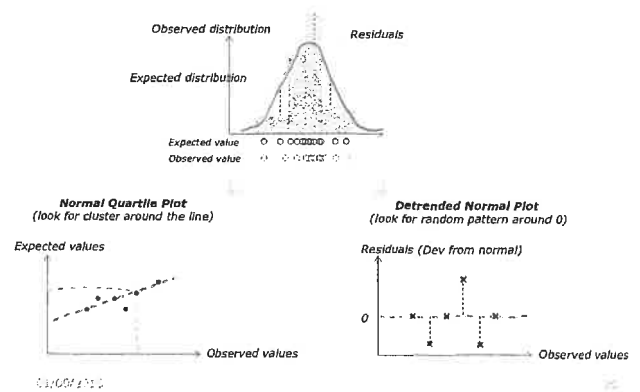
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Requirement of using F

1. Dependent variable is scale data
2. Independence
 - Randomly and independently selected sample
 - Plot values of each group in sequence to confirm there is no pattern
3. Normality of dependent variable
 - Use Histograms to check normal distribution of sample
 - Use Normal Quartile Plot to plot observed values against expected normal values
 - Use Detrended Normal Plot to plot the residuals against observed values
 - Use Formal Test of Normality to test Null Hypothesis of normal population
4. Equal variances
 - Similar within-group variances for all groups
 - Use Levene Test to test null hypothesis of equal group variances

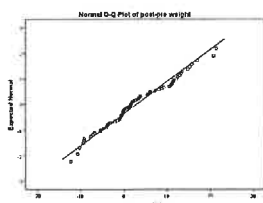


Check Normality

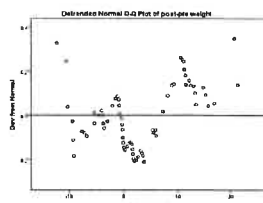


Task 6a: Check Assumption: Normality

- **Q-Q plots** of residual weight: check normality in all groups
1. Analyze > descriptive statistics > explore
 2. Set **Dependent list**: difwt
 3. > **plots** > check normality plots with tests > continue > ok



Points should cluster around the straight line for normal distributed data



Points should fall randomly around the zero line for normal distributed data

Task 6a: Check Assumption: Normality (Con't)

- 2 formal normality tests in one table:

	Tests of Normality					
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
post-pre weight	.096	72	.096	.975	72	.154

a. Lilliefors Significance Correction

If significant level is small (<0.05), the data are not in normal distribution

- Even if the normality cannot be satisfied, it is still alright to use ANOVA if the group sizes are equal.

Task 6b-1: Check Assumption: Equal variances

- **Levene test**: check assumption of equal variances in all groups
1. Analyze > Compare Means > One-way ANOVA
 2. Set **Dependent List**: difwt, set **Factor** : group
 3. > **Option** > set **Statistics** to Homogeneity of variance test > Continue > OK

Test of Homogeneity of Variances				
post-pre weight	Levene Statistic	df1	df2	Sig.
	.314	2	69	.731

Significant level = .731 means insignificant different in group variance. You may confirm the insignificant difference by looking into the variances of the groups.

(Significant level > 0.05 indicates insignificant different)

accept equal variances

Task 6b-2: Welch and Brown-Forsythe Tests

In case the assumption on homogeneity of variance (Levene test) fails, use the adjusted tests:

1. Analyze > Compare Means > One-way ANOVA
2. Set **Dependent List**: difwt, set **Factor** : group
3. **Options** > select both Brown-Forsythe and Welch > Continue > Ok

Robust Tests of Equality of Means				
post-pre weight	Statistic ^a	df1	df2	Sig.
Welch	5.355	2	41.136	.009
Brown-Forsythe	5.497	2	62.959	.006

a. Asymptotically F distributed.

Task 7a: Two-way ANOVA Test

- To run Two-way ANOVA test:
 - Use **Analyze > General Linear Model > Univariate**
 - Make settings (see red color text below)
 - Options > set Display to Descriptive statistics, Estimates of effect size, Observed Power > OK**

Variable A (Dependent List)

Variable B, C, (Fixed Factors)

	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power
Corrected Model	998992.892 ^a	5	199798.578	2.116	.070	.097	10.589	.978
Intercept	2350.408 ^a	1	2350.408	4487.392	.000	.979	4487.392	1.000
gender	316.984	1	316.984	1.426	.233	.014	1.425	.219
section	1262.726	2	631.363	2.842	.053	.054	5.694	.546
gender * section	880.444	2	440.222	2.162	.124	.042	4.319	.433
Error	21693.336	68	319.019					
Total	1086376.630	105						
Corrected Total	24343.214	104						

a. R Squared = .037 (Adjusted R Squared = .031)
b. Computed using alpha = .05

Two-way ANOVA Test (Con't)

- Two-way ANOVA Table
 - Which are the main effects?
 - Which is the interaction effect?
 - Are the effects on the test points significant?

Tests of Between-Subjects Effects

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power
Corrected Model	998992.892 ^a	5	199798.578	2.116	.070	.097	10.589	.978
Intercept	2350.408 ^a	1	2350.408	4487.392	.000	.979	4487.392	1.000
gender	316.984	1	316.984	1.426	.233	.014	1.425	.219
section	1262.726	2	631.363	2.842	.053	.054	5.694	.546
gender * section	880.444	2	440.222	2.162	.124	.042	4.319	.433
Error	21693.336	68	319.019					
Total	1086376.630	105						
Corrected Total	24343.214	104						

a. R Squared = .037 (Adjusted R Squared = .031)
b. Computed using alpha = .05

No significant effect for gender
Marginally significant effect for section
No significant gender by section interaction

Two-way ANOVA Test (Con't)

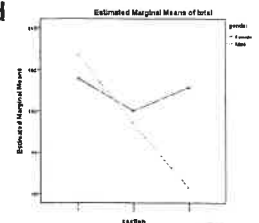
- Effect Size (Partial Eta Squared)
 - It indicates how much of the total variance is explained by each independent variable or the interaction.
- Observed Power
 - The probability of finding a significant difference at 0.05 level with the sample size of the data.
- Post-hoc Analysis
 - The procedure and interpretation of post-hoc analysis for both two-way and one-way ANOVA are the same, but only one comparison can be made.

Two-way ANOVA Test (Con't)

To visually inspect the distribution of the means:

- Statistic Tool Bar > Dialog Recall > Univariate**
- Plots > set Horizontal Axis to section > set Separate Lines to sex > Add**
- Continue > Ok**

It further confirms no significant effects by the independent variables.



Check Assumptions for Two-way ANOVA

- The assumptions are the same as One-way ANOVA. What are they?
 - Independent sample, Normality, Constant Variance
- For normality, use Q-Q Plots.**
- For constant variance (Levene's test):**
- Statistic Tool Bar > Dialog Recall > Univariate**
 - Options > Display > check Homogeneity test**
 - Continue > Ok**

Assignment 1

- Data file:** "gss.sav"
- It is the data from General Social Survey in the US. We are looking into the relationship between years of education (educ), and gender (sex) and full-time/non-fulltime employment (wrkstat = 1 and wrkstat ~= 1)
- Requirements:
 - Process and analyze the data.
 - Report and discuss the process of analysis, the results and findings.
 - Take and validate assumptions, if necessary.